

ISE 555: Traceability

21/09/2026

Contents

1	Scope and source coverage	2
2	Problem 1 — Statement and corrections	2
3	Problem 2 — Indices, units and domains	2
4	Problem 5 — Inventory conservation	3
5	Verification and presentation	3
6	Presentation revision — iteration 009	3
7	Iteration 010: author, notation and supplied exercise 6	3
8	Iteration 011: separate nutrient bounds	4

Assignment 1 — Review record

1 Scope and source coverage

All six images in raw/assignments/assignment1 belong to one document pair. IMG_7828–7830 contain the printed assignment. IMG_7897, IMG_7898 and IMG_7899 contain handwritten solutions to Problems 1, 2 and 5, respectively. The submitted material therefore supports these three selected problems. The printed instructions require at least three of the first five problems. Problems 3 and 4 have no handwritten solutions in this folder and were not newly solved. The user explicitly excluded Problem 6: no writing response or outline was created. The printed instructions also require Problem 6, submission via Brightspace by 4:00 pm Monday September 21, and typed solutions (LaTeX suggested). This document is the mathematical portion requested by the user, not a claim that the entire assignment including the mandatory writing component is complete. No submission was performed. The exercise 6 prompt is preserved in the original image only.

2 Problem 1 — Statement and corrections

Statement: IMG_7828. Solution: IMG_7897. The table correctly updates forging capacity to 90000 hours (20 times 4500), but the subsequent handwritten forging inequality appears to use 50000. The clean solution uses 90000 consistently. Drilling, milling, lathe and grinding capacities are 13500, 27000, 22500 and 45000 hours.

The original objective is written "Max $Z = x_1 = x_2$ ". This was made precise as maximize x_1 subject to $x_1 = x_2$, retaining x_1 for shafts and x_2 for frames. The coefficients 0.6/0.8, 0/0.2, 0/0.3, 0.3/0 and 0.4/0.6 were checked against the printed problem. Nonnegativity is retained. Ball bearings are purchased; unlimited availability and no separate assembly bottleneck follow the absence of such limits in the statement and are stated within the solution.

The annual maximum of 45000 assemblies and resource substitution check were added as verification of the formulation, not claimed as handwritten results. Grinding provides a matching upper bound. The requested model remains a continuous LP.

3 Problem 2 — Indices, units and domains

Statement: IMG_7828 continuing at the top of IMG_7829. Solution: IMG_7898. Costs 12, 5 and 8 dollars/bushel; total $x_1 + x_2 + x_3 = N$; four lower and four upper nutrient limits. The manuscript correctly emphasizes multiplication of each per-bushel limit by N . This explanation is retained inside the exercise.

The manuscript's upper calcium and carbohydrate rows appear to use x_3 with the limestone coefficient c_{22}/c_{23} . They are regularized to $c_{22}x_2$ and $c_{23}x_2$, respectively, as required by the ingredient definitions. In the vitamin upper row the subscript of the written upper limit appears inconsistent; it is u_4 , not u_1 . Upper bounds use the printed notation u_j throughout. Added explicit $x_1, x_2, x_3 \geq 0$, missing from the visible solution. Compressed each matching pair into a double inequality, preserving all eight constraints. No numerical coefficients or optimum were invented for the unspecified nutrient data.

4 Problem 5 — Inventory conservation

Statement: IMG_7829 continuing at the top of IMG_7830. Solution: IMG_7899. Verified demand (4000,5000,6000,4500), $I_0=2000$, costs (5,5,5.5,5.5), holding cost 1 and $I_4=0$. The handwritten objective includes I_4 ; because $I_4=0$, removing that term does not change the objective.

The manuscript lists $I_0+x_1 \geq 4000$, $I_1+x_2 \geq 5000$, $I_2+x_3 \geq 6000$ and $I_3+x_4 \geq 4500$, but these do not define how I_1, I_2, I_3 arise. Replaced by conservation equalities $I_{t-1}+x_t-I_t=d_t$ and added explicit nonnegativity of inventory as well as production. Without those links, inventory variables can create fictitious stock.

Added a concrete check: production (2000,5000,6000,4500), all ending inventories zero and objective 92750 dollars. Earlier production plus carrying cost is more expensive than on-time production under the given data. The model has no invented production capacities, sales decisions or salvage values.

5 Verification and presentation

Every numerical datum in Problems 1, 2 and 5 was compared with the photographed statements, not only with the handwriting. Statements are faithful editorial restatements including the relevant data, not verbatim transcriptions. Numerical LP checks for 1 and 5 are stored in `numerical_check.json`. Problem 2 is symbolic: its consistency is checked by indices, units and domains.

All explanations occur inside the relevant exercise. There is no final theory, general-topic or additional-information section in the assignment PDF. Figures were not added because tables and equations suffice for these formulations. The companion contains the editing history and corrections; the clean assignment contains only the selected problem statements and complete solutions. Review was performed by the assistant, not asserted to be an independent human review.

6 Presentation revision — iteration 009

At the user's request, each selected exercise occupies its own page. The assignment has no separate cover or contents page and no solution/check/interpretation labels. Numerical optima and extended verification commentary have been removed from the assignment text; their results remain in this record and `numerical_check.json`. The requested LP formulations, data, variable definitions and brief unit/balance explanations remain within their respective exercises. Problem 6 remains excluded.

7 Iteration 010: author, notation and supplied exercise 6

The user now supplies and authorizes exercise 6, superseding its earlier exclusion. The supplied English text is reproduced without rewriting; no long dashes. Author: Juan Lopez Olivan. Date: 21/09/2026. Exercises 1,2,5,6 share one document. Exercise 5 uses the handwritten production-cost notation C_{I_t} , explicitly defined as a fixed cost per gallon, not inventory. Demand D_t , holding cost, initial/final stock and both sets of decision variables are explicit. Correct inventory balances are retained. Larger separation distinguishes statements from formulations. Previous exclusion statements above describe historical iterations only.

8 Iteration 011: separate nutrient bounds

At the user's request, Problem 2 displays each lower nutrient bound and each upper nutrient bound on separate lines: four $\geq P_j$ N and four $\leq u_j$ N. The mathematical model and all coefficients are unchanged.